IQRA University (IU)		
Faculty of Engineering Sciences and Technology (FEST)		
BS Artificial Intelligence (Department of Computing and Technology (C&T))		
Course Code	**Course Name**	**Credit Hr**
CMC112	**Object Oriented Programming**	**3+1**

1. Basic Information			
Instructor	Abdul Baqi Malik	**Designation**	Lecturer
Prerequisite(s)	CMC 111	**Semester**	Fall 2026
Email	abdul.baqi@iqrauni.edu.pk	**Phone**	051-9247408
Consulting Hours	10:00-12:00 (Monday)	**Office Location**	4th Floor

2. Course Objective(s)

To understand the principles of object-oriented paradigm and introduce objects & their relationships including associations and inheritance, class hierarchy, operations on objects and use these concepts to build object-oriented solutions for real world problems.

3. Course Contents

Introduction to object oriented design, history and advantages of object oriented design, introduction to object oriented programming concepts, classes, objects, data encapsulation, constructors, destructors, access modifiers, const vs non-const functions, static data members & functions, function overloading, operator overloading, identification of classes and their relationships, composition, aggregation, inheritance, multiple inheritance, polymorphism, abstract classes and interfaces, generic programming concepts, function & class templates, standard template library, object streams, data and object serialization using object streams, exception handling

4. Course Learning Outcomes

CLOs	CLO Statement	BT Level	Mapping			% Weight
			GAs	ACM KA	SGDs	
CLO1	**Understand** the basic techniques of an object-oriented paradigm.	C2	GA2	#14 PL	9	30%
CLO2	**Examine** the objects & their relationships to build a solution for a given problem using object-oriented principles	C3	GA2			35%
CLO3	**Optimize** an object-oriented solution	C4	GA4			35%
Note: On successful completion of course GA1 (Academic Education) will automatically attain.						

5. Course Textbook / Reference Books and Supplementary Reading Material

#	Book Title	Author(s)	Edition/ publication year/publisher
1	Java: How to Program	Paul Deitel	11th Edition, 2018
2	Effective Java	Joshua Bloch	3rd Edition, 2021
3	Java for Dummies	Barry Burd	7th Edition, 2024

6. CLO Outcome Based Assessment (OBA) Tentative						
Assessment Tool		**CLO Mapped**	**CLO Marks**	**% Weight**	**Total Marks**	**Assessment Date**
Quizzes	Quiz #1	CLO 1	3	30%	3	TBD
	Quiz #2	CLO 2	3	30%	3	TBD
10	Quiz #3	CLO 3	4	40%	4	TBD
	Total Quizzes %			**100%**	10	
Assignments	Assignment #1	CLO 1	5	33%	5	TBD
	Assignment #2	CLO 2	5	33%	5	TBD
15	Assignment #3	CLO 3	5	33%	5	TBD
	Total Assignments %			**100%**	15	
Midterm	Midterm Q#1	CLO 1				TBD
	Midterm Q#2	CLO 1				TBD
	Midterm Q#3	CLO 1				TBD
25	Midterm Q#4	CLO 2				TBD
	Midterm Q#5	CLO 2				TBD
	Total Midterm %			**100%**	**25**	
Project/CCP 10	Project/CCP	CLO3	10	100%	10	TBD
	Total Project /CCP %			**100%**		
Final Exam	Final Q #1	CLO1				TBD
	Final Q#2	CLO2				TBD
	Final Q #3	CLO2				TBD
40	Final Q#4	CLO3				TBD
	Final Q#5	CLO3				TBD
	Total Final Exam %			**100%**	**40**	
100	**Total Marks**				**100**	
Note: Please make sure every CLO must be assessed at least 3 time.						

7. Weekly Plan				
Week	**Lecture No**	**Topic Covered**	**CLO**	**Assessment Tool**
1	1	Introduction to object-oriented design: History of Programming, Machine Language,	**1**	
		Assembly Language, High Level Language (Procedural Programming), Advantages of Object-Oriented Approach	**1**	
	2	Introducing Java: Features, Java basics, first java program, JRE, JDK, JVM	**1**	
2	3	Introduction to object-oriented programming concepts: Class, Objects, Object Modeling	**1**	
	4	General Form of a Class, Declaring Objects, Relation between Class and Object	**1**	
		Writing program using classes and objects, Constructors, Types of Constructors	**2**	

3	5	Advance OOP Concepts: Reference Type & Reference Variable	**1**	
	6	Primitive Types vs. Reference Types Variables, Java classes	**1**	
		Instance Variables and Methods, Static Variables and Methods	**3**	
4	7	Methods: Member Method of Class, Object as Parameter, Argument Types	**1**	
		Passing Objects to Methods, returning objects from methods	**2**	
	8	Data Encapsulation & Data hiding	**1**	
5	9	Method Overloading: Overloading methods and Constructors	**1**	
	10	Method Overloading, Message Passing Between Objects, Obtaining Handles on Objects	**1**	
		Relationships: Identification of classes and their relationships, composition, aggregation	**2**	
6	11	Inheritance: Inheritance Basics, Superclass & subclass	**1**	
		Multilevel Hierarchy, Super keyword, Multilevel Hierarchy	**1**	
	12	Polymorphism: Compile-time Polymorphism	**1**	
7	13	Polymorphism: Run time polymorphism	**2**	
		Examples of Polymorphism	**3**	
	14	**Project / CCP Discussion and Group allocation**	**3**	
8	**Midterm Exam**			
9	15	Abstract Class: Abstract methods, Final class	**1**	
	16	Interfaces: Multiple inheritance in Java, Defining an Interface, Implementing Interfaces	**2**	
		Abstract Classes vs Interfaces	**3**	
10	17	Introduction to GUI: JavaFX basics	**1**	
	18	Stage, Scene, Scene Graphs	**2**	
		Controls in JavaFX, Layouts, Graphs etc.	**2**	
11	19	Exception Handling: Exception handling basics, exception types, uncaught exceptions, using try and catch	**2**	
	20	finally block, multiple catch clauses, nested try statements	**2**	
		throw clause, some useful exceptions of java class library	**2**	
12	21	Collections: Ordered lists, Dictionaries, Sets	**1**	
	22	predefined Java collection types	**1**	
		Inventing new collections	**3**	
13	23	Generics: Generic class, Generic interfaces, Generic methods	**1**	
	24	Casting using generics, raw type	**1**	
		Wildcards & types, restrictions on generics	**3**	
14	25	Serialization: File Processing & I/O operations	**1**	
		ObjectInputStream, ObjectOutputStream	**1**	
	26	Serializable interface, Serializing & Deserializing an Object	**2**	
15	27	Inner Classes and Lambdas	**1**	
		Lambda Expressions in Java, Functional Interfaces	**3**	
	28	Using an anonymous inner class for an event listener. writing a sorting method with a lambda expression.	**3**	
16	29	Final review of all topics covered. Revision	**1/2/3**	
	30	**Project / CCP presentations and submissions.**	**3**	
17	**Final Exam**			

8. IU Assessment / grading Policy		Instructor grading for course *
Quizzes	10-15%	10%
Assignments	10-15%	15%
Projects/Presentation/CCP	0-10%	10%
Mid Semester Examination/	20-30%	25%
End Semester Examination	40-50%	40%